Karthik **Thrikkadeeri**

PhD Student, Montaño Lab

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Ecology PhD student with extensive experience in open science, quantitative workflow and project management, and code reproducibility. Proven track record of establishing ***version-controlled analytical frameworks*** for ***large-scale collaborative programmes*** (Bird Count India, State of India's Birds), developing R packages, and contributing directly to open science advancement. Strong ***attention to detail across mediums***, and passionate about written communication, code clarity, and ***transparency and accessibility*** in research.

Core Skills

Programming

- Fluent in R (advanced data wrangling, statistical ecology, package development), RMarkdown & Quarto (manuscripts, `xaringan`/`Reveal.js` slide decks, websites), Git & GitHub
- Basic experience with Shell scripting for automation, and with CSS and HTML in conjunction with RMarkdown/Quarto
- Cursory experience with Python and Julia syntax

Open-source & code standards

- Project-based workflows, script parameterisation & functionisation, version control, code annotation & documentation
- Code review in ecology research, testing and validation of package code

Editorial & communication

- Academic peer-review, citizen science data quality review
- Technical copy editing & proofreading
- Workshop instruction, coding mentoring

Relevant Experience

Coding in research

Package development

- **`skimmr` package:** Developed and maintained an open-source R package, with code testing and validation, to streamline eBird data workflows and make data manipulation accessible for researchers in the Indian context

State of India's Birds 2023

- Served as Senior Research Assistant at Nature Conservation Foundation, India (2023–24)
- Developed modular pipeline framework—unifying complex data wrangling, statistical, and visualisation scripts—for the multi-institutional collaborative project, SoIB 2023
- Parameterised and functionalised legacy scripts, adding detailed documentation and annotations to improve long-term usability and accessibility for internal teams and external collaborators
- Structured, cleaned, and organised GitHub repositories for public archiving to meet rigorous open-science standards

Bird Count India

- Served as Research Assistant at Nature Conservation Foundation, India (2021–23)
- Established comprehensive version-controlled workflows across several Bird Count India analyses, to enforce reproducibility and transparency across multi-researcher and multi-timeline projects
- Automated periodic data processing pipelines and internal metric tracking
- Created public-facing Shiny app to generate live summaries via the eBird API

Editorial & review

- **State of India's Birds 2023:** Led copy editing, proofreading and bibliography management for the public-facing technical report

- **Conference proceedings:** Managed editorial workflows, proofreading, typesetting, and publishing for academic conference proceedings at Christ University, Bengaluru, India
- **Academic peer-review:** Provided constructive peer-review evaluations for academic journals (e.g., *Journal of Threatened Taxa*, *Indian BIRDS*), with special focus on methodology, data integrity, analytical robustness, and reproducibility (2025–present)
- **Citizen science:** Contributed to improving eBird India data quality by serving as Regional Reviewer and Hotspot Editor for India (2022–present)

Advancing open science

- **BES Data and Code Hackathon Group:** Co-authored and contributed to recent *Methods in Ecology and Evolution* publication evaluating data- and code-archiving practices across BES journals, establishing recommendations for future methodological improvements
- **Instructor, GitHub workshop:** Led practical workshop introducing members of Montaño Lab at LSU to project-based workflows, repository structures, Git basics, and version control integration in everyday research pipelines
- **Instructor, R & eBird data analysis workshops:** Designed and led hands-on training sessions at Kerala Bird Monitoring Meets (2023, 2024), guiding researchers and citizen scientists through clean coding practices, eBird (big) data wrangling, and reproducible analysis in R
- **Coding mentor, LSU LAGNIAppE Programme:** Mentored post-baccalaureate scholar in R programming, script optimisation, and version control (2025–26)

Education

PhD, Biological Sciences (ongoing)

Baton Rouge, US

Louisiana State University

2025–present

- Division: Systematics, Ecology and Evolution
- Supervisor: Flavia A Montaño-Centellas

MSc, Ecology

České Budějovice, CZ

University of South Bohemia

2019–21

- Thesis: Habitat selection in post-breeding temperate forest birds
- Supervisor: Kateřina Sam

BSc, Biotechnology, Chemistry, Zoology

Bengaluru, IN

Christ University

2016–19

Selected Publications

1. N Cooper & **the BES Data and Code Hackathon Group** (2026). Data- and code-archiving in the British Ecological Society journals: Present status and recommendations for future improvements. *Methods in Ecology and Evolution* 17(7). DOI: https://doi.org/10.1111/2041-210x.70338
2. A Viswanathan, **K Thrikkadeeri**, et al. (2025). State of India's Birds 2023: A framework to leverage semi-structured citizen science for bird conservation. *Ecosphere* 16(7). DOI: https://doi.org/10.1002/ecs2.70290
3. **K Thrikkadeeri** (2024). *skimmr: Working with eBird data made easy*. R package version 0.0.0.9008. URL: https://rikudoukarthik.github.io/skimmr/
4. **SoIB** (2023). *State of India's Birds, 2023: Range, trends, and conservation status*. The SoIB Partnership. Pp. 119. DOI: https://doi.org/10.5281/zenodo.11124590